

FIRST SEMESTER DIPLOMA EXAMINATION IN MANAGEMENT

FOUNDATIONS IN SCIENCE AND TECHNOLOGY

MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A**I. Answer all questions in one word or one sentence. Each question carries one mark.****(9 x 1 = 9 Marks)**

1	State the term hypothesis in scientific method	M1.01	R
2	Write any two applications of optical fibre.	M1.04	U
3	Define law and model.	M1.01	R
4	Define solution.	M2 01	R
5	Define pH.	M2 02	R
6	State Newton's third law of motion.	M3.03	R
7	Define velocity and speed.	M3.01	R
8	Which are the fundamental particles of an atom ?	M4 01	U
9	What do you mean by atomic number ?	M4 01	U

PART B**II. Answer any eight questions from the following. Each question carries 3 marks****(8 x 3 = 24 Marks)**

1	Write the applications of LASER.	M1.04	U
2	Differentiate between inductive logic and deductive logic.	M1.01	R
3	Illustrate method of science by explaining preparation of food by plants.	M1.01	R
4	Define Molarity, Normality and Parts Per Million.	M2 01	R
5	Draw the flowchart of potable water for municipal supply.	M2 04	U

6	List the fundamental physical quantities with their units.	M3.01	R
7	Write the equations of motion in upward motion and downward motion.	M3.02	A
8	Write a short note about structure of nucleus.	M4.01	U
9	Explain mass defect.	M4 01	U
10	Write a note about different types of radio activity	M4 02	U

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain the steps in scientific method. OR	M1.01	R
IV	Explain the nature of scientific knowledge.	M1.02	U
V	Derive the relationship between pH, pOH and pK _w . OR	M2.02	A
VI	Explain buffer solution and its classification with example.	M2.02	A
VII	Write a note about hard water and soft water. How will you remove permanent hardness by ion exchange method ? OR	M2.03	U
VIII	24 ml solution of HNO ₃ neutralises 20 ml of a decinormal NaOH solution. Calculate the amount of HNO ₃ in 600 ml solution.	M2.02	A
IX	State Newton's second law of motion and derive an expression for force from it. OR	M3.03	A
X	A body moving with a velocity 20 m/s is brought to rest in 5s. Find the distance traveled before it comes to rest?	M3.02	A
XI	State and explain law of conservation of momentum.	M3.03	A

	OR		
XII	Explain recoil of gun and derive an expression for recoil velocity of gun.	M3.03	A
XIII	Distinguish between nuclear fission and nuclear fusion reaction.	M4.02	U
	OR		
XIV	Write a note about nuclear hazards and nuclear holocaust.	M4.03	U

FIRST SEMESTER DIPLOMA EXAMINATION IN MANAGEMENT

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MODEL QUESTION PAPER – SET-2

Time: 3 hours

Maximum Marks: 75

PART A**I. Answer all questions in one word or one sentence. Each question carries one mark.****(9 x 1 = 9 Marks)**

1	Define inductive logic and deductive logic.	M1.01	R
2	Write any two applications of LASER.	M1.04	U
3	State the term hypothesis in scientific method.	M1.01	R
4	Define molarity.	M2.01	R
5	What do you mean by soft water ?	M2.03	U
6	State Newton's first law of motion.	M3.03	R
7	What is momentum? Give its SI unit.	M3.03	U
8	Give an example for isotopes.	M4.01	U
9	What do you mean by mass number ?	M4.01	U

PART B**II. Answer any eight questions from the following. Each question carries 3 marks****(8 x 3 = 24 Marks)**

1	Write the applications of optical fibre.	M1.04	U
2	Explain the terms Law and model in scientific method with examples.	M1.01	R
3	Explain rocket propulsion.	M1.04	U
4	Calculate the pH of the following a) 0.1N HCl b) 0.02 N NaOH	M2.02	A
5	Write a note about standard solution, titration and end point.	M2.02	A

6	Find the dimensional formula of the following a) Force b) Power c) Pressure	M3.01	U
7	Distinguish between velocity and speed.	M3.01	U
8	Briefly explain the term nuclear energy.	M4.01	R
9	Explain nuclear fission with an example	M4.02	U
10	Name three fundamental particles of atom and mention their charges.	M4.01	R

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Explain the nature of scientific knowledge. OR	M1.02	U
IV	Explain the steps in scientific method.	M1.01	R
V	Define normality. Calculate the normality of NaOH solution containing 96 g NaOH in 600ml solution. OR	M2.01	A
VI	Define sterilisation. Explain three methods of sterilisation.	M2.04	U
VII	Distinguish between hard water and soft water. OR	M2.03	U
VIII	Explain buffer solutions with examples.	M2.02	A
IX	Explain acceleration due to gravity and write the equations of motions in upward motion and downward motion. OR	M3.02	A
X	State Newton's second law of motion and derive an expression for force from it.	M3.03	A

XI	Write a short note on different system of units. Explain SI system and its advantages. OR	M3.01	U
XII	A force of 40 N acts for 4 seconds on a body of mass 20 kg at rest. Find the velocity acquired by the body.	M3.02	A
XIII	Briefly explain harmful effects of nuclear radiations. OR	M4.03	U
XIV	Explain radio activity	M4.02	U